

Assignment 2

Adaptive Rotation Strategy

& Strategy Enhancement

Implement the Adaptive Rotation Strategy within FinRL-Trading,

then enhance it by integrating your Assignment 1 QQQ rolling logic.

ASSIGNMENT	TRACK	DUE DATE	DELIVERABLES
02 / 02	Quant Strategy	April 20, 2026	Code + Backtest + Blog

GITHUB REPOSITORY

github.com/AI4Finance-Foundation/FinRL-Trading

01 Objectives

GOAL ONE

Implement Adaptive Rotation Strategy

Carefully study and implement the Adaptive Rotation Strategy within the FinRL-Trading pipeline. Understand how it combines regime-based group selection, intra-group ranking, timing logic, and layered risk management.

GOAL TWO

Improve the Strategy

After reproducing the original strategy, integrate your QQQ rolling selection logic from Assignment 1 and explore additional signals, ranking methods, or risk controls to improve performance and robustness.

02 Assignment Tracks

TRACK A Reproduce Original Strategy

Implement the Adaptive Rotation Strategy as faithfully as possible. Explain how asset groups are defined, how market regimes affect group selection, how assets are selected within each group, and how timing and risk management layers are applied.

Focus areas:

Regime detection · Info Ratio selection · Residual momentum · Risk overlay

TRACK B Extend & Improve the Strategy

Integrate your Assignment 1 QQQ rolling selection logic to enhance the stock selection step. Compare the original strategy, your modified version, and QQQ benchmark. Explore further improvements if capable.

Focus areas:

QQQ rolling logic · Alpha signals · Regime detection · Risk controls

Key principle: A usable trading system combines asset universe design, selection logic, allocation logic, timing adjustment, risk overlay, and walk-forward evaluation. The Adaptive Rotation Strategy is a complete case study in modular strategy design.

03 Strategy Overview

The Adaptive Rotation Strategy is a research-grade, walk-forward-safe rotation framework built on the insight that different asset classes outperform under different macroeconomic regimes. Rather than committing to a fixed universe, the strategy dynamically shifts capital across four pre-defined asset groups based on recent relative performance measured by Information Ratio. Within the winning group, residual momentum selects the two strongest individual assets, while timing and risk-control layers manage drawdowns in real time.

4

Asset Groups

12w

IR Lookback

Top 2

Final Picks

Weekly

Rebalance

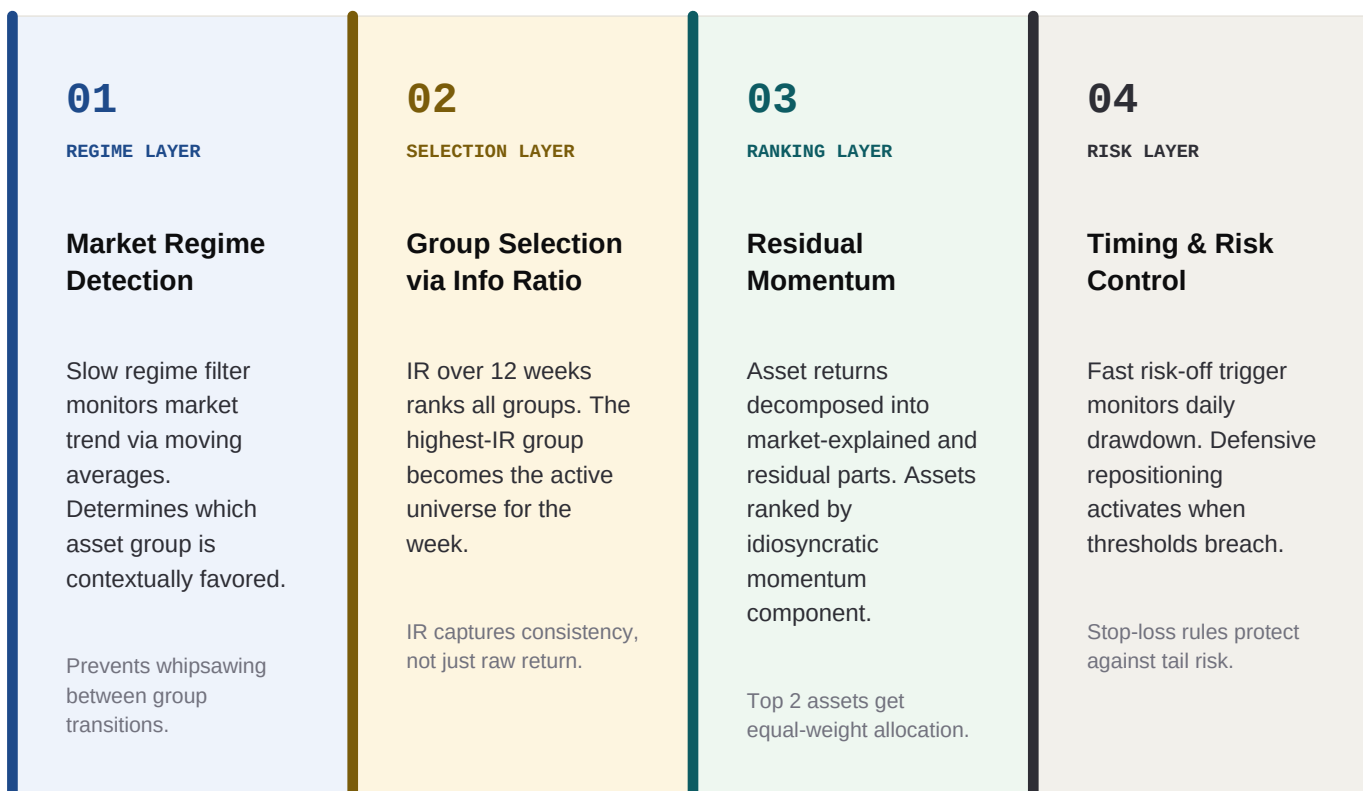
3.1 Asset Groups

GROWTH TECH	REAL ASSETS	DEFENSIVE	DEFAULT
Momentum-driven; outperforms in risk-on bull markets	Inflation hedge; favored in commodity/energy cycles	Capital preservation; active in risk-off regimes	Broad market fallback when no regime is dominant
AAPL MSFT NVDA META AMZN GOOGL TSLA	XOM CVX FCX BHP GLD SLV	TLT IEF XLU XLV IAU SHY UUP	SPY QQQ IAU XLU XLV

3.2 Execution Logic

GROUP SELECTION	INTRA-GROUP RANK	REBALANCE FREQ	FINAL PICKS
Information Ratio (12-week) IR = $\text{mean}(\text{excess return}) / \text{std}(\text{excess return})$ computed over trailing 12 weeks for each group.	Residual Momentum Regress asset returns on market factor; rank by residual (idiosyncratic) momentum component.	Weekly Portfolio is rebalanced every week; position weights are updated based on latest rankings.	Top 2 per group Equal-weight allocation to the top 2 assets in the selected group each period.

3.3 Four-Layer Pipeline



3.4 Research Questions

- Does QQQ rolling selection improve the Growth Tech sleeve performance?
- Does the modified strategy improve Sharpe ratio or reduce max drawdown?
- Is the improvement stable across time, or only in certain market regimes?

- Which layer matters most: group selection, intra-group ranking, or risk control?
- What trade-off exists between upside capture and downside protection?
- If your strategy performs better, what mechanism explains the difference?

04 What You Should Do

1

Understand the Strategy Mechanism

Before coding, explain the full pipeline: why multiple asset groups are used, why Information Ratio drives group selection, why residual momentum ranks within groups, and how timing/risk layers affect performance.

2

Implement the Adaptive Rotation Strategy

Reproduce the major components — group selection via IR, intra-group ranking via residual momentum, weekly rebalancing, and risk management layers.

3

Replace Stock Selection with Assignment 1 Logic

Integrate your QQQ rolling selection strategy from Assignment 1: replace the Growth Tech universe with a rolling-selected subset, use it as a pre-filter, or introduce it as an alternative candidate pool for comparison.

4

Evaluate Whether the New Combination Performs Better

Compare the original strategy, your modified version, and QQQ benchmark. Report cumulative return, annualized return, Sharpe ratio, max drawdown, and volatility. Additional benchmarks (SPY, equal-weight) are welcome.

5

Optional Strategy Enhancements [optional]

Explore new alpha signals, alternative regime indicators, improved momentum ranking, modified rebalance frequency, volatility targeting, or cash allocation.

05 Deliverables

A	B	C
CODE IMPLEMENTATION Implement both the original and modified Adaptive Rotation Strategy in the FinRL-Trading pipeline. Code should be clean and reproducible. original strategy · modified version	BACKTEST RESULTS Equity curves and key metrics for original, modified, and QQQ benchmark. Include Sharpe ratio, max drawdown, annualized return, and volatility. equity curves · vs. QQQ	MEDIUM BLOG Strategy understanding, implementation logic, backtest results with QQQ comparison, and analysis of why the strategy works or does not work. strategy analysis · insights

06 Evaluation Criteria

#	Criterion	Description	Weight
01	Correctness of Implementation	Did you correctly reproduce the main logic of the Adaptive Rotation Strategy?	
02	Quality of Modification	Did you meaningfully integrate Assignment 1 ideas rather than superficial changes?	
03	Depth of Analysis	Did you clearly explain the mechanism and performance of the strategy?	
04	Experimental Rigor	Did you compare against proper benchmarks and present results clearly?	
05	Original Thinking	Did you explore reasonable improvements, extensions, or new alpha ideas?	